



Multiscale temperature-dependent ceramic matrix composite damage model with thermal residual stresses and manufacturing-induced damage

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ABSTRACT

This work presents a multiscale thermomechanical simulation framework to capture the temperature-dependent damage behavior of woven ceramic matrix composites (CMCs). The framework consists of cool-down simulations, which capture a realistic material initial state, and subsequent mechanical loading simulations to capture temperature-dependent nonlinear stress-strain behavior. The cooldown simulations result in a realistic material initial state with thermal residual stresses and damage hotspots that occur due to constituent property mismatch and post-manufacturing cooldown. A fracture mechanics-informed thermomechanical progressive damage model is extended to capture the manufacturing-induced damage that occurs because of the high thermal residual stresses and to simulate the mechanical response of two-dimensional (2D) plain weave carbon (C) fiber, silicon carbide (SiC) matrix (C/SiC) CMCs at temperatures ranging from room temperature (RT) to 1200 °C. A combination of temperature-dependent material properties and damage model parameters are included in the model to simulate the effects of temperature on deformation and damage behavior. Model calibration was conducted using quasi-static tensile experimental data from the literature for RT, 700 °C, and 1200 °C and the nonlinear, temperature-dependent predictive capabilities of the reformulated model are demonstrated for 1000 °C. The model is also applied to simulate the temperature-dependent thermomechanical response of a 2D woven five harness satin (5HS) SiC/SiC CMC at RT and 1200 °C and shows excellent agreement with experiments.

1. Introduction

Ceramic matrix composites (CMC) are emerging as materials of choice for components operating in extreme service conditions such as those seen in the propulsion and energy production fields [1–3]. The relatively low density, wide range of operating temperatures, and desirable properties of CMCs, including strength, stiffness, toughness, and damage resistance [4–6], have positioned CMCs as ideal replacements for conventional high temperature materials such as nickel-based superalloys. However, CMCs exhibit complex temperature-dependent thermomechanical deformation and damage behavior that occurs across multiple length scales. This complex multiscale behavior requires any predictive analysis framework to consider scale-specific features and behaviors and incorporate them in a synergistic manner. A multiscale modeling framework that can capture the scale-specific and temperature-dependent damage behavior using

physics-based constitutive models is required to accurately predict CMC damage behavior in service environments.

During their service life, CMCs are exposed to a wide range of operating temperatures, which has a significant effect on the CMC response [7,8]. Accounting for these thermal effects is critical to understanding CMC damage behavior and material performance under a wide array of operating conditions. The manufacturing process also plays an important role in CMC response and damage. Most SiC matrix CMCs are manufactured using chemical vapor infiltration (CVI), polymer infiltration and pyrolysis (PIP), or melt infiltration (MI), all of which occur at extremely high temperatures (up to ~1200 °C for CVI, up to ~1600 °C for PIP, and up to ~1400 °C for MI) [9,10]. As a result of spatially and architecturally varying thermal properties, CMCs experience significant thermal residual stresses when they are cooled from processing to operating temperatures. Appreciable as-produced damage can occur as a result of the high thermal residual stresses and must be accounted for to accurately capture the initial material state. Typi-

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cal C/SiC CMCs are manufactured using an initial CVI step to deposit a thin interfacial pyrolytic carbon (PyC) layer on the fiber surfaces, followed by deposition of the SiC matrix using CVI [9]. Similarly, an initial CVI step is also used to deposit a thin interfacial boron nitride (BN) layer on the fiber surfaces of most SiC/SiC CMCs. The subsequent matrix densification process for SiC/SiC CMCs typically consists of an initial CVI matrix deposition step followed by MI to fill voids and achieve higher matrix density [10]. This results in two distinct SiC phases as well as pockets of excess Si in the SiC matrix of CVI/MI SiC/SiC material systems. Each phase has distinct thermomechanical properties which must be accounted for to accurately model the overall weave thermomechanical damage response. In general, C/SiC CMCs are more susceptible to manufacturing-induced damage than SiC/SiC CMCs because of the higher mismatch between constituent thermal properties. Additionally, the relatively high-density matrix resulting from the two-step matrix densification process used to manufacture SiC/SiC CMCs is more resistant to matrix cracking and manufacturing-induced damage because the MI step fills voids and cracks left by the CVI step.

Most CMC models assume a pristine initial state, or at most assume initial degraded material properties to match the stiffness of the as-produced CMC. Manufacturing-induced damage can result in reduced material performance that must be accounted for to accurately model CMC thermomechanical deformation and damage behavior. Yang et al. [11] studied the thermomechanical degradation and microstructural evolution of oxide/oxide CMCs due to cyclic thermal shocks. They showed that differences in constituent thermal properties resulted in significant microporosity formation and material property degradation when rapidly cooled. In addition to the microporosity, they observed delamination due to high temperature gradients, and matrix cracking due to in-plane thermal tensile stresses. The high cooling rate exacerbated these cooldown-induced damage mechanisms, but thermal property mismatch can result in significant damage and property degradation even when cooled gradually, and should be included in CMC models. Gawayed et al. [7] investigated the effects of temperature on the elastic properties of a SiC/SiC CMC and correlated the global weave elastic properties to the in situ temperature-dependent properties of the constituent phases. This is a valuable work that increases understanding of the temperature-dependent response of SiC/SiC CMCs, but it fails to account for thermal residual stresses and manufacturing induced damage. Yang et al. [8] accounted for thermal residual stresses due to cooldown in a 2D plain weave C/SiC CMC and were able to model the as-produced matrix cracks with a reasonably accurate initial modulus.

Damage in CMCs is temperature dependent and occurs across multiple length scales. Damage behavior in brittle matrix CMCs is typically dominated by matrix damage mechanisms, particularly in monotonic tensile tests of brittle matrix CMCs with higher matrix modulus than fiber modulus [8]. In these CMCs, the matrix stress increases more rapidly than the fiber stress and damage initiates in the matrix in the form of tow level matrix cracks followed by intratow weave level matrix cracking. These damage mechanisms eventually progress and lead to fiber/matrix debonding followed by fiber fracture and pullout at high strains. Longbiao [12] studied the damage and fracture behavior of a SiC fiber, magnesium aluminosilicate (MAS) matrix CMC under thermomechanical fatigue loading at different temperatures. They showed that significant material degradation occurs at high temperatures due to an increase in matrix microcracking as well as oxidation and weakening of the fiber/matrix interphase. Ladeveze et al. [13] studied room temperature damage in SiC/SiC CMCs and developed thermodynamic constitutive damage models to capture the CMC stress-strain response. Arhatari et al. [14] studied damage mechanisms in C/C-SiC CMCs under four-point bending using X-ray radiography. They observed scale-dependent damage mechanisms including intertow and intratow matrix microcracks, fiber-matrix

interfacial debonding, progressive fiber filament failure, fiber pullout, and delamination.

The scale-dependent and temperature-dependent CMC damage behavior necessitates the use of multiscale thermomechanical modeling methods to accurately capture temperature and damage effects at each length scale. The multiscale generalized method of cells (MSGMC) developed by Liu et al. [15] enables concurrent analysis of an arbitrary number of length scales and is well suited to capture scale-dependent material behavior. In this case, the CMC weave is divided into three main length scales—the microscale (fiber/matrix level), mesoscale (fiber tows and intertow matrix) and macroscale (full composite weave). The global thermomechanical behavior of the CMC weaves can be obtained by simulating each relevant length scale and applying appropriate material constitutive models. Borkowski et al. [16] used MSGMC to investigate the effects of cooldown and simulate the as-produced damage state of RT plain weave and UD C/SiC CMCs. By accounting for thermal residual stresses, they were able to capture a realistic initial state with local concentrations of damage in the RT UD CMC. In a recent publication, the authors developed a matrix damage model using fracture mechanics and crack growth kinetics and implemented the model using MSGMC to capture the effects of damage in RT C/SiC CMCs [17]. The model effectively captured mechanisms of RT brittle matrix damage but failed to account for thermal residual stresses and manufacturing-induced damage.

In this work, a multiscale thermomechanical simulation framework is developed and implemented using MSGMC to capture the temperature-dependent damage behavior of woven CMCs. The framework consists of cooldown simulations, which capture a realistic material initial state, and subsequent mechanical loading simulations to capture the temperature-dependent nonlinear stress-strain behavior. A fracture mechanics-informed thermomechanical progressive damage model previously developed by the authors [17] is reformulated and extended to capture manufacturing-induced damage and to simulate the temperature-dependent mechanical response of 2D plain weave CVI C/SiC CMCs at temperatures ranging from RT to 1200 °C. Additionally, a secant-tangent matrix damage model [18] is employed to capture the homogenized effects of local subcell failure and simulate the post-cracking matrix subcell behavior. Model calibration was conducted using experimental data from the literature for RT, 700 °C, and 1200 °C. The experimental data for the 1000 °C case was set aside for validation purposes and to evaluate the predictive capabilities of the model. Experimental data from the 1000 °C case was not used to calibrate model parameters. The model agrees well with experiments and can effectively predict the CMC response within the temperature range studied in this work. With minor modification, the model can simulate the temperature-dependent response of nearly any woven brittle-matrix composite. To demonstrate this capability, the model has been applied to simulate the temperature-dependent thermomechanical response of a 2D woven 5HS CVI/MI SiC/SiC CMC and shows excellent agreement with experiments. The Curtin progressive fiber filament failure model [19], which has been adapted for use in GMC-based models [20], is used in this work to capture the stochastic nature of progressive fiber failure in SiC/SiC CMCs.

2. Model description

2.1. Multiscale modeling framework

The temperature dependent matrix damage model is applied to the cooldown simulations and subsequent mechanical loading simulations using the MSGMC framework. This methodology enables prediction of a realistic residual stress profile with corresponding as-produced damage for various use temperatures and permits accurate temperature-dependent simulations to capture the matrix damage behavior of various woven CMC material systems across a wide temperature range.

The relevant length scales (micro-, meso-, macroscales) for a 2D woven CMC are indicated in Fig. 1, which shows an idealized 5HS triply periodic weave representative unit cell (RUC) that is discretized into subcells. The simulated global strain is applied at the macroscale level to the weave RUC, and the subcell strains are obtained using localization. Each subcell of the macroscale RUC is then modeled using a doubly periodic mesoscale RUC which is in turn discretized into subcells. Additional localization is again performed to obtain the mesoscale RUC subcell strains. The subcells of the mesoscale RUC are modeled using a microscale RUC consisting of SiC matrix, C or SiC fiber (monofilament), and PyC or BN fiber coating (interphase) subcells. A final localization is performed to determine the strain of each monolithic constituent subcell, and the stress–strain constitutive relationship corresponding to the material is applied to determine stress.

The stresses at each length scale are determined using homogenization techniques. For the microscale RUCs, which are the subcells of the mesoscale RUC, the homogenized stress is obtained by volume averaging the stresses of each monolithic constituent subcell. This method results in the microscale RUC effective stiffness, which includes damage effects and depends on the microscale geometry and constituent material properties. Similarly, in the mesoscale RUCs, which are subcells of the macroscale RUC, the homogenized stress is obtained by volume averaging the stresses of each microscale RUC. The mesoscale RUC effective stiffness is also computed. Finally, the global macroscale RUC stress and the global homogenized damaged composite stiffness matrix are obtained by volume averaging of the mesoscale RUCs. These localization/homogenization algorithms allow for an accurate semi-analytical solution that links scale-specific deformation and damage behavior to the global weave response.

2.2. Cooldown simulations

The CVI matrix densification process commonly used to manufacture CMCs requires extremely high temperatures. By accounting for spatially and architecturally varying thermal properties, the framework developed in this work can capture realistic post-cooldown thermal residual stress profiles and initial damage states for woven CMCs. In this paper, 2D plain weave C/SiC and 5HS SiC/SiC CMC material systems are simulated. The C/SiC material system modeled in this work consists of T300 carbon fibers with a PyC interphase material and CVI SiC matrix. The SiC/SiC material system consists of Sylramic-iBN fibers with a BN interphase material and CVI SiC matrix at the tow level and MI SiC matrix at the weave level. The CTE for the T300 carbon fibers and CVI and MI SiC matrix materials are shown in Fig. 2, and the temperature-dependent Young's modulus for the CVI and MI matrix are shown in Fig. 3. The temperature-dependent material properties for the PyC and BN interphases as well as for the Sylramic-iBN fibers are included in Table 1. The T300 fibers are modeled as transversely isotropic, and the CVI and MI SiC matrix as well as the PyC and BN interphase materials and the Sylramic-iBN SiC fibers are modeled as thermally and elastically isotropic. The T300 fiber elastic properties, which are assumed to be approximately constant over the temperature domain, are included in Table 2.

The simulated woven composite architecture is assembled as described in the previous section, and cooldown simulations are conducted to determine the thermal residual stresses. The processing temperatures, which are used as the beginning temperatures for the cooldown simulations, for the plain weave C/SiC material and the 5HS SiC/SiC material are assumed to be ~1200 °C [9] and ~1400 °C [10], respectively. The actual cooldown rates are unpublished by the manufacturers, but gradual cooldown is assumed and simulated in increments of 1–4 °C. The low cooldown rate implies that no significant thermal gradients occur due to the slow passive cooling. The damage model, which will be discussed in the following section, is applied to determine the as-produced damage in the weave due to the high thermal residual stresses before subsequent mechanical loading simu-

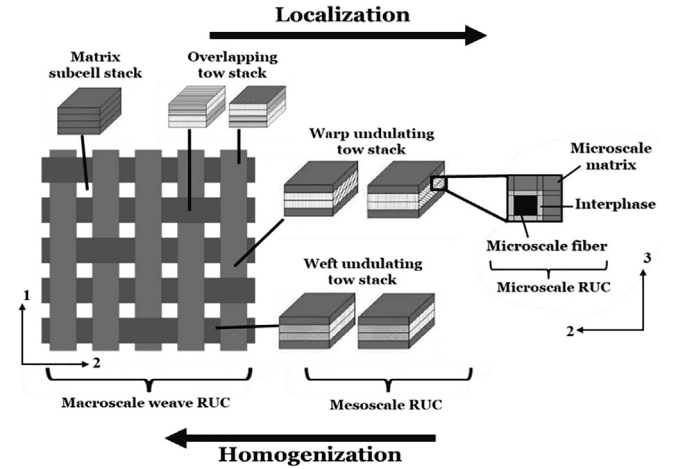


Fig. 1. Schematic of MSGMC multiscale modeling methodology.

lations at various temperatures are applied. By capturing realistic thermal residual stress profiles and initial damage states, this methodology allows more accurate simulations of temperature-dependent damaged CMC stress–strain behavior.

2.3. Damage model

The matrix material stress–strain constitutive relationships are obtained as discussed in [17] by taking the Helmholtz free energy as the scalar state potential of the material and assuming that it is completely described by the linear thermoelastic strain potential. The scalar state potential governs the evolution of state variables, which can be chosen to represent specific deformation and damage mechanisms. A unified isotropic damage internal state variable (ISV), D , is chosen to represent the damaged state of the material and includes effects of micropore nucleation and growth as well as matrix crack activation and propagation. The unified damage ISV is used in the following damaged constitutive relationship obtained by applying the principle of strain equivalence for a damaged medium and by taking partial derivatives:

$$\sigma_{ij} = \rho \frac{\partial \psi}{\partial \varepsilon_{ij}} = C_{ijkl} \varepsilon_{kl}^e (1 - D), \quad (1)$$

where $i, j, k, l = 1, 2, 3$, σ_{ij} is a component of the Cauchy stress tensor, ρ is the density, ψ is the Helmholtz free energy (see [17] for full definition), ε_{ij} is a component of the total strain tensor, C_{ijkl} is a component of the forth-order stiffness tensor, and ε_{kl}^e is a component of the elastic portion of the strain tensor. The evolution of the unified damage ISV is defined as a piecewise function to represent the effects of two separate damage mechanisms:

$$\dot{D} = \begin{cases} \dot{D}_p K_I \leq K_{IC} \\ \dot{D}_c K_I > K_{IC} \end{cases}, \quad (2)$$

where K_I and K_{IC} are the mode I stress intensity factor (SIF) and critical SIF (fracture toughness), respectively. Due to the brittle nature of the ceramic matrix material, the mode II and III SIFs play a negligible role in the rate of crack growth [28] and are not included in the model. The evolution equations for D_p , which captures effects of micropore (pre-crack) nucleation and growth, and D_c , which captures effects of matrix crack activation and propagation, are defined as follows:

$$\dot{D}_p = a(1 - D_p)\gamma \varepsilon^V \quad (3a)$$

$$\dot{D}_c = \frac{\pi^2}{10} (1 + \nu) (5 - 4\nu) \eta l^2 \dot{l} \quad (3b)$$

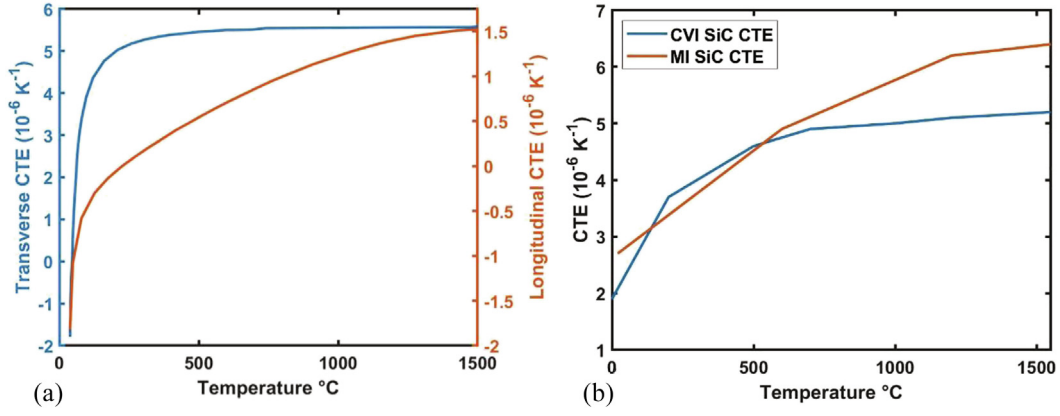


Fig. 2. CTE vs. temperature; (a) Longitudinal and transverse CTE for T300 carbon fiber [21]; (b) CTE for CVI [22] and MI [23] SiC matrix.

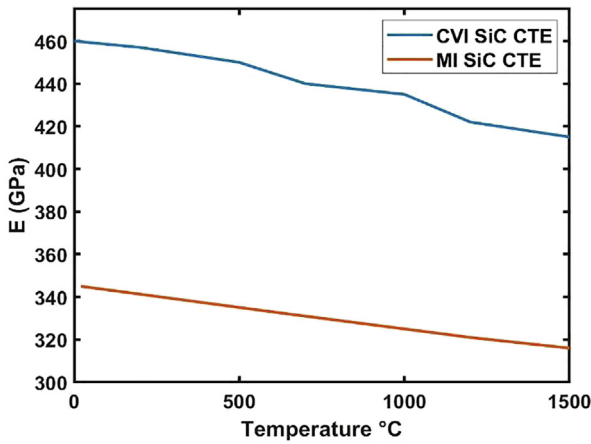


Fig. 3. Young's modulus vs. temperature for CVI [22] and MI [23] SiC matrix.

where a is a temperature-dependent scaling parameter related to microporosity density, γ is the entropy dissipation rate defined by Rubin et al. [29], ε^V is the volumetric strain, ν is the Poisson's ratio, η is a parameter related to crack density and growth rate, and l is the characteristic flaw size. Note that the parameter Γ_p , used in calculating γ as described in Ref. [29] has been modified to $\Gamma_p = \frac{3G}{\sigma_{eq}}$, where G is the shear modulus, and σ_{eq} is the equivalent stress. Additionally, note that the microporosity (precrack) density scaling parameter, a , is assigned two separate values in implementation to capture the precrack growth behavior during cooldown (a_0) as well as during mechanical loading (a_1). The evolution of l is defined as:

$$\dot{l} = \begin{cases} b\dot{D}_p K_I \leq K_{IC} \\ \frac{K_I - K_{IC}}{K_I - K_{IC}/2} K_I > K_{IC} \end{cases} \quad (4)$$

where b is a proportionality constant relating the flaw size before crack initiation to the microporosity damage variable. As shown in Eq. (4), after crack activation, the crack extension rate is computed using a

modified form of the crack growth kinetic relationship given in Paliwal et al. [30]. Defining the characteristic flaw size in this way allows the model to simulate both microporosity and cracking damage mechanisms, and the transition from micropore (precrack) to matrix crack. Initially, the matrix degradation is dominated by the nucleation and growth of microstructural pores which grow as stress increases until the SIF due to the flaw exceeds a critical value. At this point, the SIF exceeds the fracture toughness of the material, matrix cracking is activated, and the continued effects of crack propagation are simulated as flaw growth is dominated by crack growth kinetics. This methodology aligns with the physical damage nucleation and propagation behavior of brittle materials. In these materials, micropores/precracks form and then grow until reaching a critical size after which the flaw transitions to crack-like behavior and propagation [31].

Matrix cracking in real CMCs is directionally dependent and causes local anisotropy. To capture this behavior in the model, the cracked matrix is assumed to be transversely isotropic in the plane of the crack, and matrix anisotropy occurs when the damaged stiffness tensor is rotated from the crack plane to the global axes. The definition of the damage ISV in Eq. (2) is applied to obtain a transversely isotropic damaged compliance tensor, S , in the plane of the crack. In matrix form:

$$S = \begin{bmatrix} 1/E_t & -\nu/E_t & -\nu/E_n & 0 & 0 & 0 \\ -\nu/E_t & 1/E_t & -\nu/E_n & 0 & 0 & 0 \\ -\nu/E_n & -\nu/E_n & 1/E_n & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G_m & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G_m & 0 \\ 0 & 0 & 0 & 0 & 0 & 2(1+\nu)/E_t \end{bmatrix}, \quad (5)$$

where E_t is the transverse modulus ($E_t = E$), and G_m is the cracked shear modulus, which is assumed to follow the form $G_m = \frac{E_n}{2(1+\nu)}$. The Poisson's ratio in a cracked body is expected to slightly increase [32], but some researchers have observed no significant change [33] or even a decrease in Poisson ratio [34] after onset of cracking in a brittle material. Due to a lack of experimental data for this material system and the conflicting results in the literature, the Poisson's ratio is assumed to be unaffected by damage. The compliance in the global axes are obtained

Table 1

Temperature-dependent material properties for PyC and BN interphase materials and Sylramic fibers.

	CTE ($\times 10^{-6}$)		E (GPa)		ν	
	20 $^{\circ}\text{C}$	1200 $^{\circ}\text{C}$	20 $^{\circ}\text{C}$	1200 $^{\circ}\text{C}$	20 $^{\circ}\text{C}$	1200 $^{\circ}\text{C}$
PyC [24]	-1.12	1.7	20	20	0.23	0.23
BN [23,7]	6.3	3.7	20.3	14	0.22	0.22
Sylramic-iBN [25,26]	5.4	5.4	400	365	0.17	0.17

Table 2
Elastic properties for T300 carbon fibers [27]

E_1 (GPa)	E_2 (GPa)	G_{12} (GPa)	G_{23} (GPa)	ν_{12}	ν_{23}
230	15	27	7	0.3	0.35

by rotating the damaged compliance tensor from principal axes to global axes as follows (in Einstein notation):

$$S'_{ijkl} = R_{ip}R_{jq}R_{pr}R_{qs}R_{kr}R_{ls}, \quad (6)$$

where $\mathbf{R}$ is the rotation matrix whose rows are the unit vectors associated with the principal stresses and are the basis vectors for the principal frame, and the indices $i, j, \dots, r, s$ range from 1 to 3. As a result of the rotation from principal axes, the resulting compliance tensor, $\mathbf{S}'$, and the corresponding stiffness tensor, $\mathbf{C}'$, are fully anisotropic in the global frame where loads are applied

In a real CMC material system, local failure can occur, causing load redistribution to surrounding regions of the material. This local failure and load redistribution behavior is difficult to capture explicitly using MSGMC. The first order displacement fields used in MSGMC endow the method with computational efficiency, but when combined with the imposed traction continuity requirements, failure of individual subcells (decreasing subcell stiffness to zero) results in entire subcell rows and columns being overly compliant. This in turn has the potential to cause the entire RUC to become overly compliant in some directions depending on the architecture. As a result, an unrealistic decrease in the homogenized RUC stiffness at that length scale can occur, and the overall composite modulus can be unrealistically degraded. To avoid this issue, failure in individual subcells is not explicitly simulated in this work. Instead, a secant-tangent matrix damage model [18] is employed to capture the homogenized effects of local subcell failure and simulate the post-cracking matrix subcell behavior. This supplementary damage model is applied to “failed” subcells to prevent their stiffness from reaching zero to avoid overly compliant material behavior. The damaged normalized secant modulus, n , is a calibrated parameter selected to match the post-crack stiffness of the simulated CMC with experimental results.

2.4. Fiber damage model

The failure strain of T300 carbon fibers used to simulate the response of the plain weave C/SiC CMC is approximately three times that of the ultimate strain for typical C/SiC CMCs at RT [35]. Failure in these CMC material systems is typically dominated by matrix damage and failure behavior that leads to failure at the weave level, and fiber failure plays a minor role in the global composite response. As a result, fiber damage is not included in the C/SiC CMC damage model. However, a matrix damage model alone is insufficient to fully capture the SiC/SiC damage response. The Sylramic fibers in the 5HS weave SiC/SiC CMC simulated as part of this study are stoichiometric SiC fibers, and as such, have damage and failure behavior similar to that of the SiC matrix material. The Curtin progressive fiber filament failure model [19], which has been adapted for use in GMC-based models [20], is used in this work to capture the stochastic nature of progressive fiber failure in SiC/SiC CMCs. This modified Curtin model is based on fiber strength statistics combined with shear lag analysis and defines the damaged fiber modulus, E_f^* , as follows:

$$E_f^* = \frac{1}{2} \left[1 + \exp \left(- \left(\frac{\sigma_L}{\sigma_C} \right)^{m+1} \right) \right] E_f, \quad (7)$$

where E_f is the undamaged fiber modulus, σ_L is the fiber longitudinal stress, m is the fiber Weibull modulus, and σ_C is defined as:

$$\sigma_C = \left(\frac{\sigma_0^m \tau L_0}{r} \right)^{\frac{1}{m+1}}. \quad (8)$$

In this equation, σ_0 is the characteristic fiber strength, L_0 is the fiber characteristic length, r is the fiber radius, and τ is the fiber–matrix interfacial sliding resistance. The ultimate strength and failure strains predicted by the model are given by:

$$\sigma_f^{ult} = \frac{1}{2} \sigma_C h^{\left(\frac{1}{m+1}\right)} [1 + \exp(-h)], \quad (9)$$

$$\varepsilon_f^{ult} = \frac{\sigma_C}{E_f} h^{\left(\frac{1}{m+1}\right)}, \quad (10)$$

where h is the lowest positive number that satisfies:

$$1 + (1 - h(m+1)) \exp(-h) = 0. \quad (11)$$

The parameter values, σ_0 , L_0 , r , and m , were obtained from [36] and are 2.1 GPa, 25.5 mm, 4.5 μm , and 5, respectively. The fiber–matrix interfacial sliding resistance parameter, τ , was assumed to be 10 MPa. For the 1200 °C case, τ and m were assumed to be 1 MPa and 2, respectively, based on the high-temperature fiber data in [37,38].

2.5. Model calibration

The damage model parameters used to simulate the scale-dependent matrix damage behavior in this work are obtained by calibrating with experimental results from the literature. Yang et al. [8] performed experiments across a range of temperatures using dogbone 2D plain woven C/SiC specimens machined from a CMC panel. The longitudinal tensile experiments were performed under displacement control at a quasistatic strain rate of $3 \times 10^{-5} \text{ s}^{-1}$. The simulations discussed in this paper were displacement controlled at a comparable quasistatic strain rate of $1 \times 10^{-5} \text{ s}^{-1}$. Gawayed et al. [7] performed longitudinal tensile tests using flat 2D 5HS weave SiC/SiC specimens per EPM testing standards, which are equivalent to ASTM C1359, at RT and 1204 °C. The experimental results from both studies are included in Fig. 4, which clearly shows the proportional elastic limit (PEL) for the high temperature C/SiC load cases and both SiC/SiC load cases. The PEL corresponds to the onset of first macroscale matrix cracking [18] and marks the transition between local, tow-level damage to intratow weave level damage. The RT C/SiC CMC load case has no obvious PEL because the high manufacturing-induced damage causes significant matrix cracking that rapidly extends under load and degrades material properties almost immediately. The PEL increases with temperature for the C/SiC CMC, despite a reduction in constituent properties with increasing temperature. This is primarily because the residual stresses and manufacturing-induced damage become less significant with increasing temperatures. The SiC/SiC CMC has less mismatch in constituent thermal properties, so the manufacturing-induced damage is less severe. The higher constituent properties at RT combined with relatively low manufacturing-induced damage results in a higher PEL at RT than at high temperatures for SiC/SiC.

A critical objective of this work is to capture a realistic initial damage state resulting from thermal residual stresses due to cooldown from extreme processing temperatures. Micrographs of as-produced C/SiC CMCs (see Fig. 5) show significant manufacturing-induced damage in the form of matrix cracking and transverse tow cracks. These initial manufacturing-induced damage mechanisms occur in specific

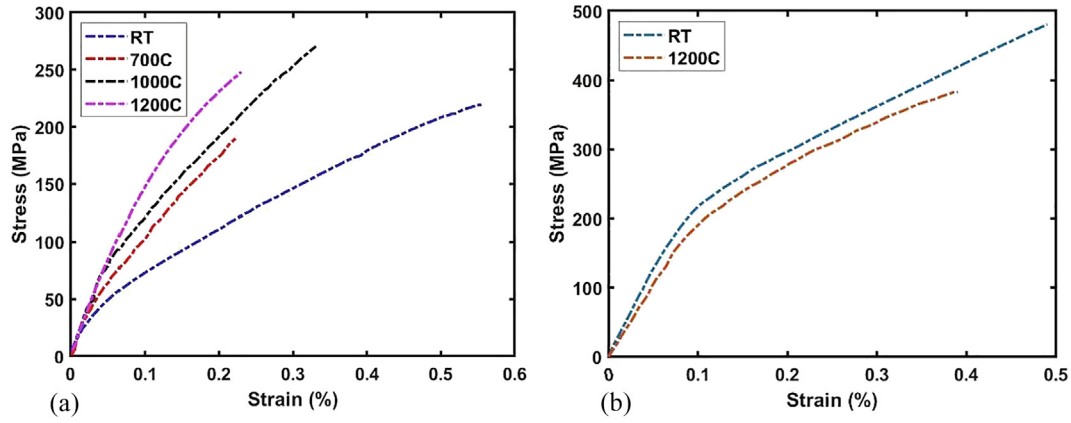


Fig. 4. Experimental longitudinal stress–strain response. (a) 2D plain weave C/SiC CMC [8]; (b) 2D SHS weave SiC/SiC CMC [7]

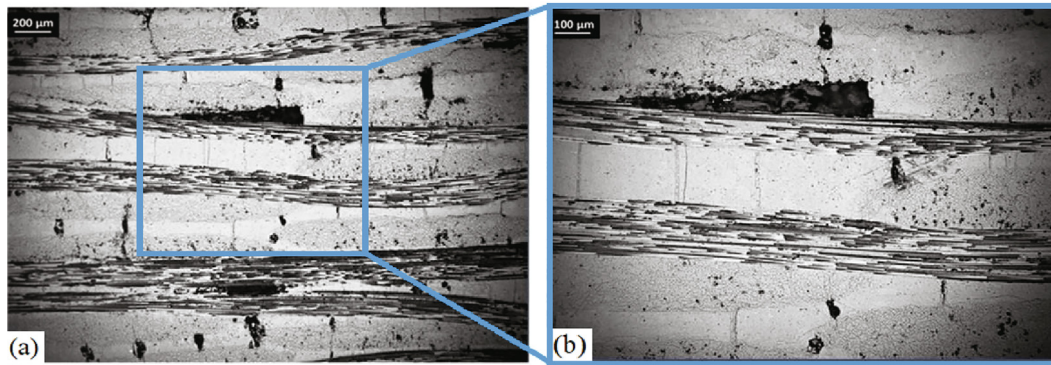


Fig. 5. Micrographs adapted from [39] showing as-produced damage with preferential orientation in CVI C/SiC CMC. (a) Partial specimen cross section; (b) Close-up view.

areas within the CMC weaves and show preferential orientations. The location and orientation of these as-produced cracks are governed by the thermal residual stress profiles which are themselves governed by the weave architecture and constituent thermal properties. The cooldown simulation framework described in this work accounts for these factors and captures realistic thermal residual stress profiles. Corresponding damage hotspots occur in regions consistent with those shown in Fig. 5. This will be discussed in greater detail in the following sections.

The CVI/MI manufacturing method used to fabricate the SiC/SiC specimens simulated in this work results in a more dense matrix (see Fig. 6) with less porosity and fewer open voids in the weave. Additionally, the CTEs of the individual constituents in the SiC/SiC CMC have significantly less mismatch compared to those of the C/SiC CMC. As a result, the SiC/SiC CMC has lower thermal residual stresses, and the CVI/MI SiC matrix material is more resistant to matrix cracking and transverse tow cracking because of thermal shrinkage. The micrograph in Fig. 6 shows significantly less initial damage compared to the C/SiC material in Fig. 5. The results from the cooldown simulation framework developed in this work agree with this experimental observation, as will be shown in later sections.

3. Results and discussion

3.1. Monolithic SiC

The stress–strain response of bulk SiC matrix material is simulated and compared to the linear elastic case to demonstrate the capabilities of the damage model. In this case, the cooldown simulations have no

effect because the monolithic SiC matrix material has no CTE mismatch, and therefore no thermal residual stresses. The quasi-brittle behavior of the SiC matrix material is captured and shown in Fig. 7. The natural cracking behavior is displayed in Fig. 8(a), which shows the value of the total damage ISV. Initially, the damage behavior is dominated by micropore/precrack nucleation and propagation, which causes slight material nonlinearity at low strains. At approximately 0.04% strain, matrix cracks activate and damage propagation is dominated by matrix cracking and crack growth kinetics. The damage variable grows rapidly after crack activation and failure occurs shortly thereafter. To satisfy the governing thermodynamic requirements inherent in ISV theory, the temporal evolution of the damage ISVs must monotonically increase and satisfy normality conditions. These conditions are met by the dissipation-based definition of D_p below $\sim 0.04\%$ strain, and by the fracture mechanics-based definition of D_c after matrix crack activation. The piecewise definition of D and its growth rates are thermodynamically valid and agree with ISV theory. The effective modulus of the monolithic SiC material is shown in Fig. 8 (b). Gradual material property degradation occurs because of the material dissipation associated with the formation and growth of micropores/precracks. After matrix crack activation, cracking propagates rapidly causing more significant material property degradation that continues until failure.

3.2. C/SiC

The initial thermal residual stresses resulting from cooldown from $\sim 1200^\circ\text{C}$ to RT in a 2D plain weave C/SiC CMC are shown in Fig. 9. The same procedure was followed for all temperature cases (except

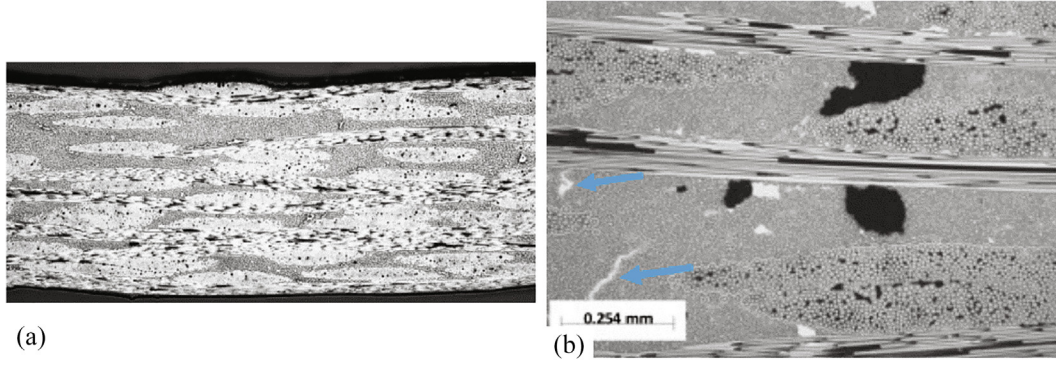


Fig. 6. Micrographs adapted from [7] showing as-produced damage in CVI/MI SiC/SiC CMC. (a) Specimen cross section; (b) Micrograph showing matrix cracks filled after MI step.

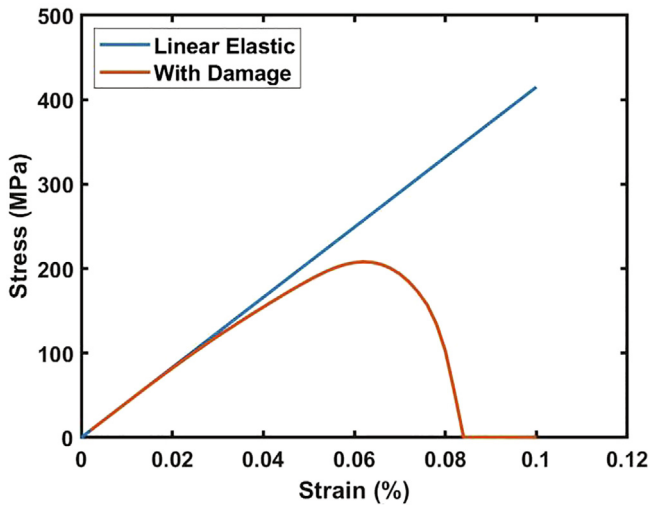


Fig. 7. Simulated stress-strain response for RT monolithic SiC.

with different final temperatures for the cooldown segment of the simulations), but only the RT case is included here for reference. The highest tensile stresses, which contribute the most to the cooldown-induced damage, occur in the transverse (weft direction) and longitudinal (warp direction) shown in Fig. 9(a) and (b), respectively. These high tensile stress areas correspond to regions in as-produced CMCs with matrix cracks and transverse tow cracks caused by material

shrinkage as seen in Fig. 5. The directionally dependent damage model is applied and results in degraded properties in high stress areas according to the direction of the maximum principal stress in each matrix subcell. This results in a realistic anisotropic initial damage state for the simulated as-produced weave that includes damage hot-spot areas. The temperature-dependent stress-strain responses (see Fig. 11) are obtained by performing subsequent mechanical loading simulations.

Model calibration was conducted using nonlinear least squares regression optimization techniques to obtain the optimum temperature dependent model parameters to simulate the response of a 2D plain weave C/SiC CMC at RT, 700 °C, and 1200 °C. The experimental data for the 1000 °C case was set aside to evaluate the predictive capabilities of the model and was not used to calibrate model parameters. The calibrated model parameters and temperature trends for each parameter are shown in Fig. 10. The temperature trends of the model parameters can provide insight into the temperature-dependent CMC deformation and damage behavior. For instance, the monotonically decreasing behavior of the cooldown and mechanical loading microporosity scaling parameters, a_0 (Fig. 10(a)) and a_1 (Fig. 10(b)), indicates that the magnitude of the precrack damage variable growth rate decreases with increasing temperature. This aligns with expectations and experimental observations because the manufacturing-induced cooldown damage is less significant at higher operating temperatures. The decreasing behavior of the proportionality constant, b (Fig. 10(c)), which relates the flaw size before crack initiation to the value of the porosity damage variable (effective size of micropores/crack precursors), indicates that the relationship between the effective size of the flaws and the value of the precracking damage ISV decreases with tem-

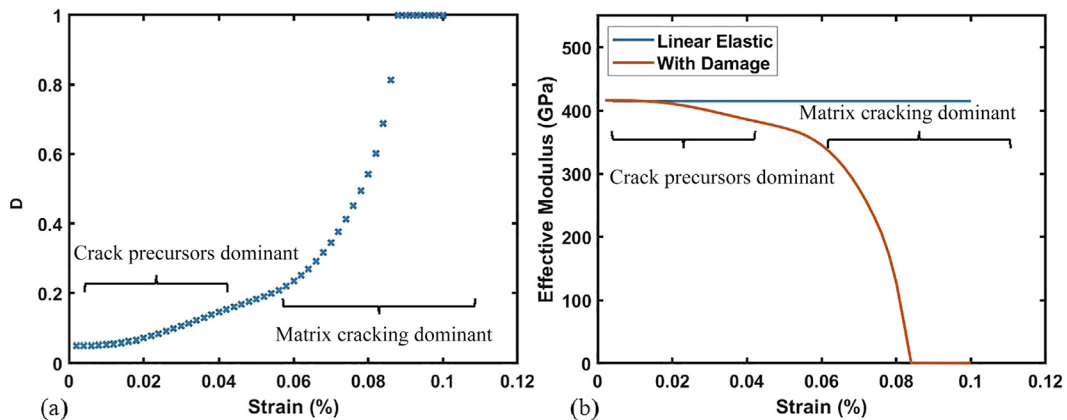


Fig. 8. Simulation results for RT monolithic SiC. (a) Total damage ISV vs. strain; (b) Effective SiC modulus vs. strain.

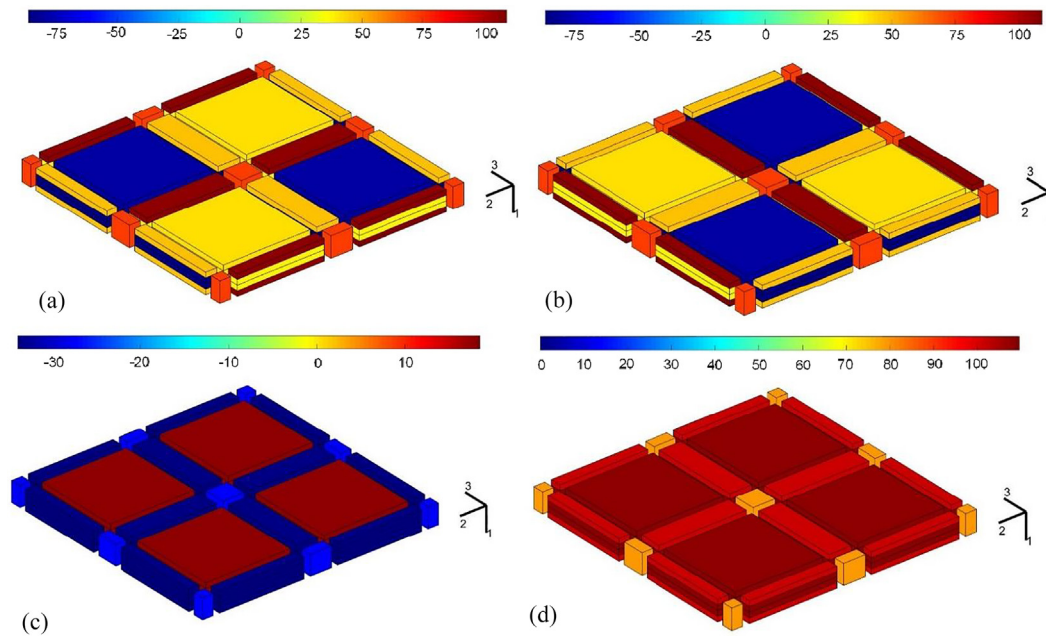


Fig. 9. Simulated thermal residual stresses for RT plain weave C/SiC CMC. (a) Residual transverse stress (weft direction); (b) Residual longitudinal stress (warp direction); (c) Residual through-thickness stress; (d) Residual effective stress.

perature. A weak monotonically increasing trend is displayed by the crack density/crack growth rate parameter, η (Fig. 10(d)), indicating that the severity of cracking and rate of crack extension will moderately increase with temperature. In this case, the effects become more pronounced with temperature, indicating that for extremely high temperatures beyond the range studied in this work, crack growth and subsequent fracture would be even more severe and would occur more rapidly. This aligns with expectations and experimental observations. A weak monotonically increasing trend is displayed by n (Fig. 10(e)), the post-crack secant-tangent modulus that ensures subcell stiffness is not reduced completely to zero to prevent overly compliant simulated material behavior. Because n is a numerical feature of the model, no physically meaningful interpretation of this parameter's temperature-dependent trends can be inferred. Unfortunately, the lack of published temperature-dependent CMC stress-strain response data limits confidence in the observed trends of each parameter. Published experimental data for equivalent material systems and loading conditions was available for four temperatures, leaving only three for model calibration and one for prediction and model validation. The authors are currently planning in-house tests to collect data at additional temperatures, which will allow more in-depth investigation into the temperature trends of each model parameter and will increase confidence in the temperature trends of the model parameters.

The multiscale cooldown and thermomechanical simulation framework and damage model can accurately represent the nonlinear stress-strain response of the CMC across a wide temperature range, as shown in Fig. 11. In this figure, the color of the curve corresponds to the temperature, solid curves represent the simulated response, and dashed curves represent experimental results. The model accurately captures the dominant damage behavior of the brittle matrix CMCs. At low strains, the damage growth is dominated by tow-level crack initiation and propagation in the undulating tows. This corresponds to the initial nonlinearity in Fig. 11. As strain increases, matrix cracking saturates in the undulating tows and cracks initiate in non-undulating tows just before the PEL (transition between tow-level damage and macroscale matrix cracking, as discussed in Section 2.5). This is reflected in the model, and matrix crack activation in the intratow

matrix corresponds to the rapid drop in stiffness at the PEL. As strain increases, matrix cracks continue to propagate in the intratow weave-level matrix subsells before reaching saturation.

3.2.1. Predictive capability

The nonlinear, temperature dependent predictive capability of the model is demonstrated in Fig. 12, which compares the predicted response at 1000 °C to the experimental response at the same temperature. The predicted damage model parameters for 1000 °C were chosen to match the temperature trends of each parameter shown in Fig. 10. With more data points (experimental results from additional temperatures), more complex relationships could be investigated to better capture the effects of temperature on each parameter. In this case, however, the lack of experimental results precludes more accurate investigation, and only simple linear fits were applied to each parameter. A linear fit is deemed appropriate for the purposes of this study. The temperature trends for each parameter within the range of RT to 1200 °C are monotonic, so the rough overall temperature-dependent behavior of each parameter within this range is retained using linear fits. However, more accurate relationships (sigmoidal or exponential trends) would be necessary to attempt to predict the CMC response outside the temperature range investigated in this work. The predicted response and experimental results for 1000 °C show good agreement, which serves to validate the predictive capabilities of the temperature dependent model within the temperature range considered in this work.

3.3. SiC/SiC

The model developed as part of this work effectively captures the quasi-brittle damage response of the SiC matrix used in C/SiC CMCs. By modifying the weave-level RUC and performing additional parameter calibration, this model can be applied to nearly any woven brittle-matrix composite. To demonstrate this capability, the model has been applied to simulate the temperature-dependent thermomechanical response of a 2D woven 5HS SiC/SiC CMC. The CVI and MI matrix densification steps used to manufacture the SiC/SiC CMC results in

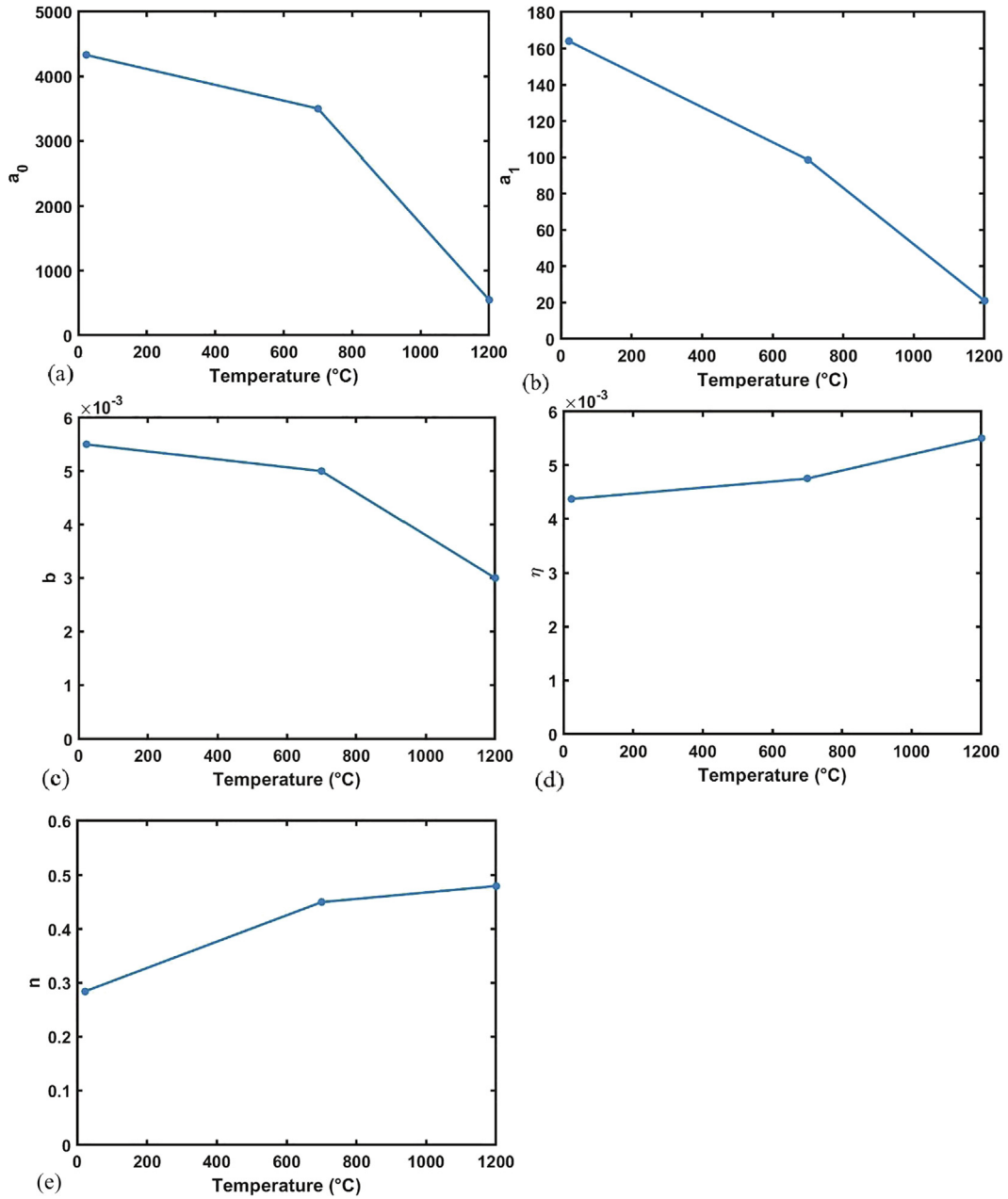


Fig. 10. Damage model parameter temperature trends. (a) Porosity scaling parameter—cooldown; (b) Porosity scaling parameter—loading; (c) Effective pore size proportionality constant; (d) Crack density/growth rate parameter; (e) Post-cracking secant-tangent damage parameter.

two SiC matrix phases that were modeled as separate materials with distinct thermomechanical properties. A small volume fraction of excess Si pockets often forms as a result from the MI step, but these were assumed to have a negligible effect on the overall weave thermomechanical damage response and were not accounted for in this work. The more complicated architecture of the 5HS weave is displayed in Fig. 13, which shows the thermal residual stresses resulting from simulated cooldown from ~1400 °C to RT. The CTE mismatch between the constituents of the SiC/SiC CMC is much less than that of the C/SiC CMC. Additionally, the CVI/MI SiC/SiC is more resistant to matrix cracking because of the higher density matrix that occurs as voids and cracks are filled during the MI step. As a result, the SiC/SiC CMCs have lower thermal residual stresses and less significant

manufacturing-induced damage despite the higher processing temperatures required for MI.

After the cooldown simulations, mechanical loading simulations are performed to obtain temperature-dependent stress strain responses (see Fig. 14), which correlate well with experiments. As before, solid curves correspond to simulation results, dashed curves represent experimental results, and the colors of the curves indicate the corresponding experimental/simulation temperature. The calibrated model parameters used to simulate the RT and 1200 °C SiC/SiC nonlinear thermomechanical response are included in Table 3. The rough trends observed in the C/SiC model appear to hold true here as well, and the less significant manufacturing-induced damage observed in the SiC/SiC material system relative to the C/SiC CMC is reflected in the low

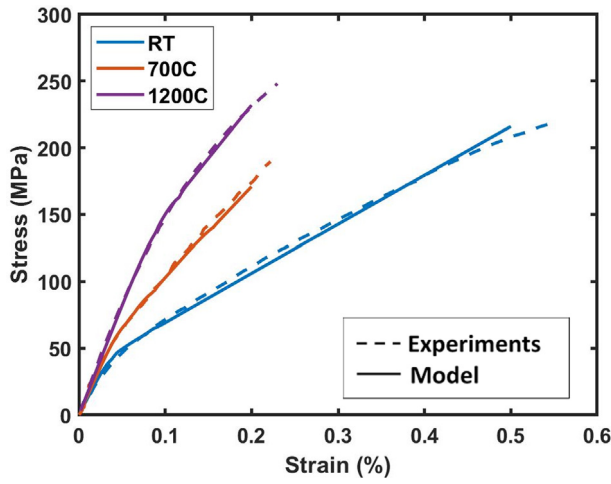


Fig. 11. Temperature-dependent model results compared with experimental data from Ref. [8].

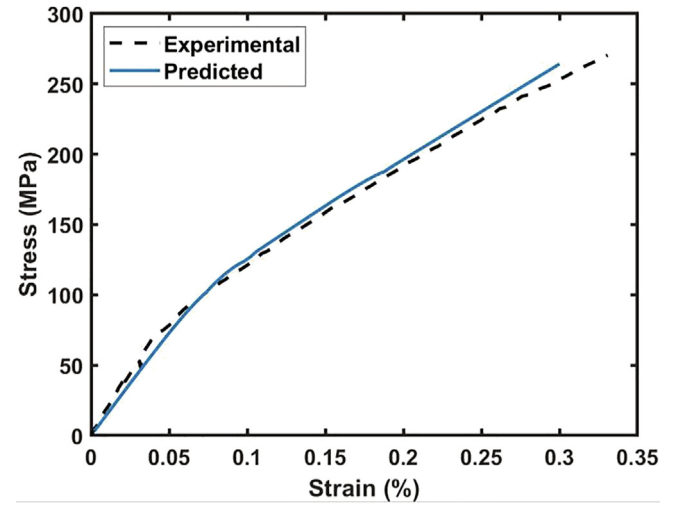


Fig. 12. Predicted C/SiC longitudinal response compared to experiment [8] for 1000 °C.

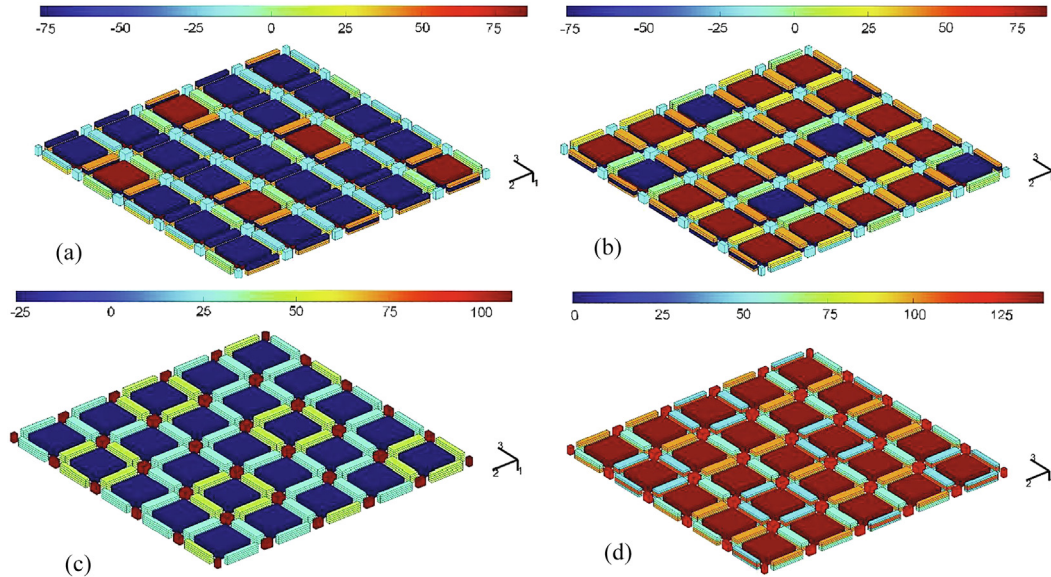


Fig. 13. Simulated thermal residual stresses for RT 5-HS weave SiC/SiC CMC. (a) Residual transverse stress (weft direction); (b) Residual longitudinal stress (warp direction); (c) Residual through-thickness stress; (d) Residual effective stress.

values of the a_0 parameter in the SiC/SiC model compared to the C/SiC model.

4. Conclusion

A multiscale thermomechanical CMC simulation framework is developed and implemented using MSGMC to capture the temperature-dependent damage behavior of woven CMCs. The modeling framework was applied to 2D woven CMCs including plain weave C/SiC CMCs manufactured using CVI and 5HS SiC/SiC CMCs manufactured using a two-step CVI/MI matrix densification process. The nonlinear predictive capabilities of the model are demonstrated and show excellent agreement with the literature. The framework consists of cooldown simulations to capture thermal residual stresses and subsequent mechanical loading simulations to capture deformation and damage behavior. A fracture mechanics-informed thermomechanical

progressive damage model is used to simulate the cooldown damage that occurs as a result of the high thermal residual stresses and to capture the temperature-dependent damage behavior of the CMCs at temperatures ranging from RT to 1200 °C. A unified damage ISV is defined that combines the effects of microporosity/precrack nucleation and growth with crack activation and propagation. Crack growth kinetics and fracture mechanics are used to determine the transition from precrack to crack as well as to simulate the effects of crack propagation on the matrix mechanical properties. The main outcomes of this work are as follows:

- Formulation of residual stresses due to manufacturing related cool down and investigation of flaws and damage in as-produced CMCs.
- Incorporation of the cool down simulation with a reformulated fracture mechanics-informed damage model for more accurate prediction of temperature-dependent damage and failure in CMCs.

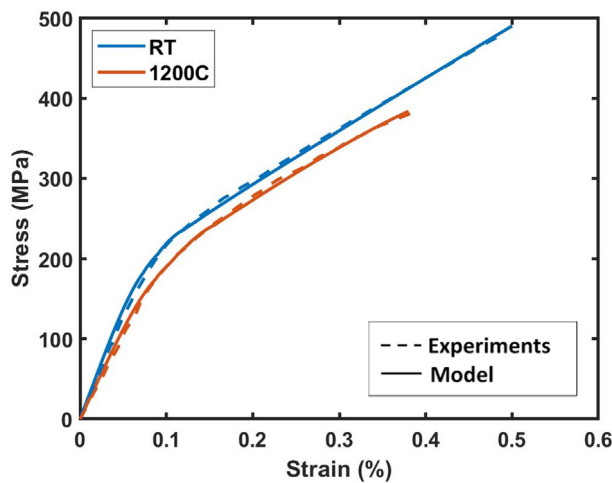


Fig. 14. Temperature-dependent SiC/SiC model results compared to experiments from Ref. [7].

Table 3
SiC/SiC matrix damage model parameters for RT and 1200 °C.

	a_0	a_1	b	η	n
RT	1100	400	2.21×10^{-5}	1.22×10^{-4}	0.14
1200 °C	520	250	1.07×10^{-5}	1.92×10^{-4}	0.18

- Implementation of the new damage model within a multi-physics thermomechanical multiscale methodology to capture CMC damage progression and deformation behavior under thermomechanical loading.

CRediT authorship contribution statement

Travis Skinner: Conceptualization, Methodology, Software, Data curation, Visualization, Investigation, Validation, Formal analysis, Writing - original draft, Investigation, Visualization. **Aditi Chattopadhyay:** Conceptualization, Project administration, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

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